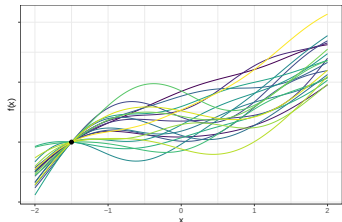
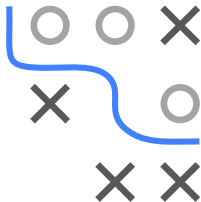


Introduction to Machine Learning

Gaussian Processes

Mean functions for GPs



Learning goals

- Trends can be modeled via specification of the mean function

ZERO-MEAN FUNCTIONS

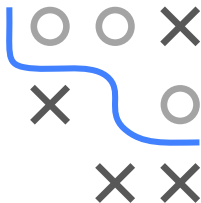
- Previously: common assumption of zero-mean prior

$$m(\mathbf{x}) \equiv 0$$

- Prior knowledge + inference solely handled via $k(\cdot, \cdot)$
- Implication: $m(\cdot)$ not relevant for posterior process

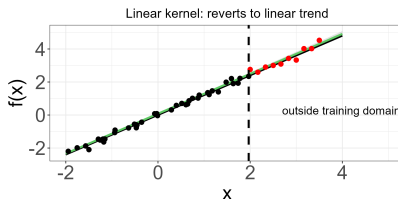
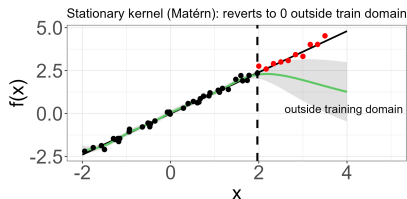
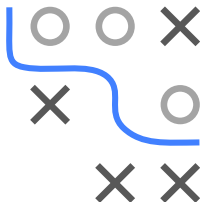
$$\mathbf{m}_{\text{post}} = \mathbb{E}(\mathbf{f}_* | \mathbf{X}_*, \mathbf{X}, \mathbf{y}_*) = \mathbf{K}_* \mathbf{K}_y^{-1} \mathbf{y}, \quad \mathbf{K}_{\text{post}} = \mathbf{K}_{**} - \mathbf{K}_*^T \mathbf{K}_y^{-1} \mathbf{K}_*$$

- Not necessarily drastic limitation: **posterior** mean generally $\neq 0$
- If data follow some trend $m(\mathbf{X})$, we can always center them by subtracting $m(\mathbf{X}) \Rightarrow \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ applicable again



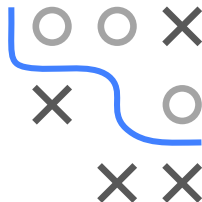
TREND VIA COVARIANCE STRUCTURE

- For zero-mean GPs with stationary kernels, posterior mean reverts to the prior further outside the training domain (no extrapolation)
- But trend-like behaviour could be directly encoded in $k(\cdot, \cdot)$:
 - Linear kernel: $k(\mathbf{x}, \mathbf{x}') = \sigma^2 \mathbf{x}^\top \mathbf{x}'$
 - Polynomial kernels for global polynomial trends
 - Composite kernels: $k = k_{\text{long}} + k_{\text{short}}$
- Produces non-reverting priors even with $m(\mathbf{x}) = 0$, but lower interpretability and kernel-dependent extrapolation
- Consider GP for DGP with linear trend:



WHY MODEL A TREND EXPLICITLY?

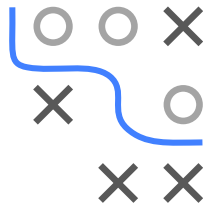
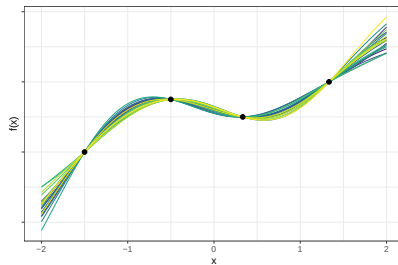
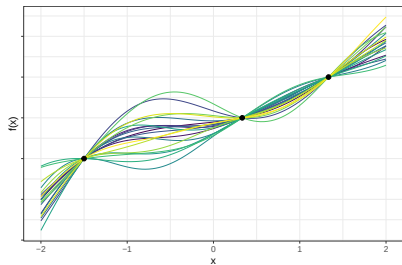
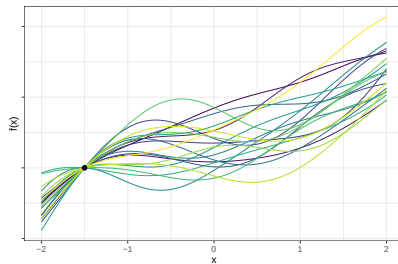
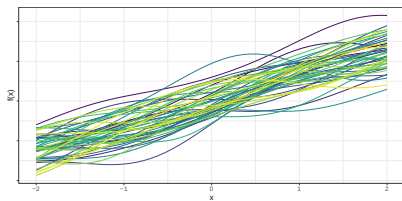
- Still: can make sense to model $m(\cdot)$ explicitly as potentially nonzero
 - **Efficiency:** kernel $k(\cdot, \cdot)$ need not mimic global structure via very long lengthscales
 - **Extrapolation:** outside data range, $\mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ reverts to flat mean
 $\Rightarrow$ often unrealistic
 - **Interpretability:** clear separation between systematic trend and stochastic fluctuations
 - **Prior knowledge:** encode known effects (linear, seasonal, additive)
- Assuming $\mathcal{GP}(m(\cdot), k(\cdot, \cdot))$, posterior mean with $m(\cdot)$ becomes
$$\mathbf{m}_{\text{post}}(\mathbf{X}_*) = m(\mathbf{X}_*) + \mathbf{K}_* \mathbf{K}_y^{-1} (\mathbf{y} - m(\mathbf{X}))$$
- Trend $m(\mathbf{X}_*)$ = interpretable global component; Correction = GP adjustment around this trend; Variance stays $= \mathbf{K}_{**} - \mathbf{K}_*^{\top} \mathbf{K}_y^{-1} \mathbf{K}_*$



NON-ZERO-MEAN FUNCTIONS

- GPs with **trend**

$$m(\mathbf{x}) = 1.5x$$

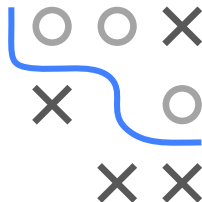


SEMI-PARAMETRIC GP

- (Deterministic) mean functions $m(\cdot)$ often hard to specify
- Solution: **semi-parametric** GPs combining global (often linear) model + zero-mean GP for residuals

$$g(\mathbf{x}) = m_{\beta}(\mathbf{x}) + f(\mathbf{x}), \quad f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$$

- In principle: **any model** $m(\cdot)$ can be used
 - Fixed parametric: $m_{\beta}(\mathbf{x}) = \beta_0 + \mathbf{x}^{\top} \beta$
 - Basis expansions: $m_{\beta}(\mathbf{x}) = \mathbf{b}(\mathbf{x})^{\top} \beta$
 - Flexible ML models: GLMs, boosting, neural nets, ...



- Log marginal likelihood:

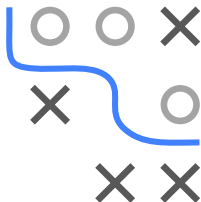
$$\ell(\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2) = -\frac{1}{2} \mathbf{r}^\top \mathbf{K}_y^{-1} \mathbf{r} - \frac{1}{2} \log |\mathbf{K}_y| - \frac{n}{2} \log(2\pi),$$

with $\mathbf{r} = \mathbf{y} - m_{\boldsymbol{\beta}}(\mathbf{X})$

- **Joint estimation:** maximize ℓ over all parameters
- **Sequential:** fit $m(\cdot)$ first, GP on residuals
⇒ ignores uncertainty from first stage, variance underestimated
- **Fully Bayesian:** priors on $(\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2)$, posterior inference via MCMC or VI

$$p(\boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2 \mid \mathbf{y}, \mathbf{X}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \boldsymbol{\theta}, \sigma^2, \mathbf{X}) p(\boldsymbol{\beta}) p(\boldsymbol{\theta}) p(\sigma^2)$$

- For complex $m(\cdot)$, estimation by full Bayesian inference or joint likelihood becomes computationally difficult



SEPARABILITY OF GRADIENTS

- Gradients of ℓ decompose neatly into:

$$\nabla_{\beta} \ell = \left(\frac{\partial m_{\beta}(\mathbf{x})}{\partial \beta} \right)^{\top} \mathbf{K}_y^{-1} \mathbf{r},$$

$$\nabla_{\theta} \ell = \frac{1}{2} \mathbf{r}^{\top} \mathbf{K}_y^{-1} \frac{\partial \mathbf{K}_y}{\partial \theta} \mathbf{K}_y^{-1} \mathbf{r} - \frac{1}{2} \text{tr} \left(\mathbf{K}_y^{-1} \frac{\partial \mathbf{K}_y}{\partial \theta} \right)$$

- Trend parameters β enter only via $\mathbf{r}$ and the design/basis functions
- Kernel hyperparameters θ and noise σ^2 enter only via $\mathbf{K}_y$ and its derivatives
- Consequence: updates are **decoupled in form**, though they interact through $\mathbf{r}$ and $\mathbf{K}_y^{-1}$

